PROJECT DEFENSE & TECHNICAL VIVA VOCE

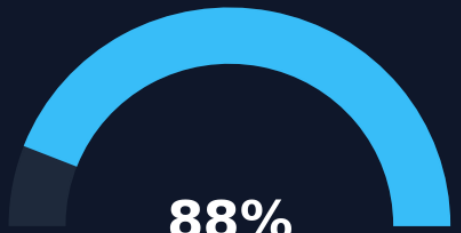
ReviewGuard AI

Advanced Fake Product Review Detection & Trust Analysis System

ACADEMIC PROJECT PROFILE

Candidate Name:	[Your Name]
Department:	Computer Science & Engineering
Project Domain:	Machine Learning & Natural Language Processing
Project Guide:	[Guide Name]
Target Audience:	120-Person Academic Viva & Project Defense
System Architecture:	Dual-Model NLP Pipeline (CART + SAMME)

OPERATIONAL TRUST TELEMETRY



Product Trust Score

Verified High Trust Level

Dual-Model Pipeline

Sentiment Decoupled

SSRF Protected Scraper

ReportLab PDF Engine

THE E-COMMERCE REVIEW MANIPULATION CRISIS

Why Traditional Star Averages Fail Against Coordinated Marketplace Fraud

ORGANIC REVIEWS

Authentic Customer Experiences

- Natural vocabulary richness (TTR ~ 0.75+)
- Balanced pros and cons (constructive)
- Specific experiential nouns & details
- Gradual, organic distribution over time
- Example: 'Battery lasts 7h, case slightly slippery.'

5-STAR ASTROTURFING

Paid Sybil Promotion (Inflating Bad Goods)

- Low lexical entropy / template copy
- Extreme emotional hyperbole without facts
- Excessive punctuation (WOW!!! BEST EVER!!!)
- Temporal burst clusters from new accounts
- Example: 'AMAZING MIRACLE PRODUCT FIVE STARS BUY!'

1-STAR REVIEW BOMBING

Competitor Smear Attacks (Damaging Good Goods)

- Hostile smear templates without product defect facts
- Targeted coordinated attack on rival merchants
- Zero verified purchase authentication
- Drives legitimate sellers out of business
- Example: 'CHEATER SCAMMER WORST TRASH DO NOT BUY!'

CORE VIVA FINDING: Traditional arithmetic averages treat all 3 categories equally. ReviewGuard AI mathematically separates them.

THE CONFLATION FALLACY: SENTIMENT VS. AUTHENTICITY

Addressing the Teacher's Core Challenge: A Good Product with Fake Reviews

QUADRANT 1: GENUINE POSITIVE

Authentic Customer Endorsement

Real buyers praising high-quality products.
• Features detailed usage facts & natural TTR.
• Example: 'Earbuds fit snugly, 7h battery life.'
• System Action: Accepted into Product Trust Index.

QUADRANT 2: FAKE POSITIVE (ASTROTURFING)

Manipulative Promotional Astroturfing

Inorganic 5-star praise on good OR bad products.
• Saturated with extreme superlatives & bot templates.
• Example: 'MIRACLE BEST PRODUCT IN HISTORY 1000%!'
• System Action: Flagged as Fake; Penalized in Trust Score.

QUADRANT 3: GENUINE NEGATIVE

Constructive Customer Defect Feedback

Real buyers reporting legitimate product defects.
• Concrete aspect complaints with balanced wording.
• Example: 'Screen arrived cracked, RMA took 3 days.'
• System Action: Retained as Genuine; Vital for seller QA.

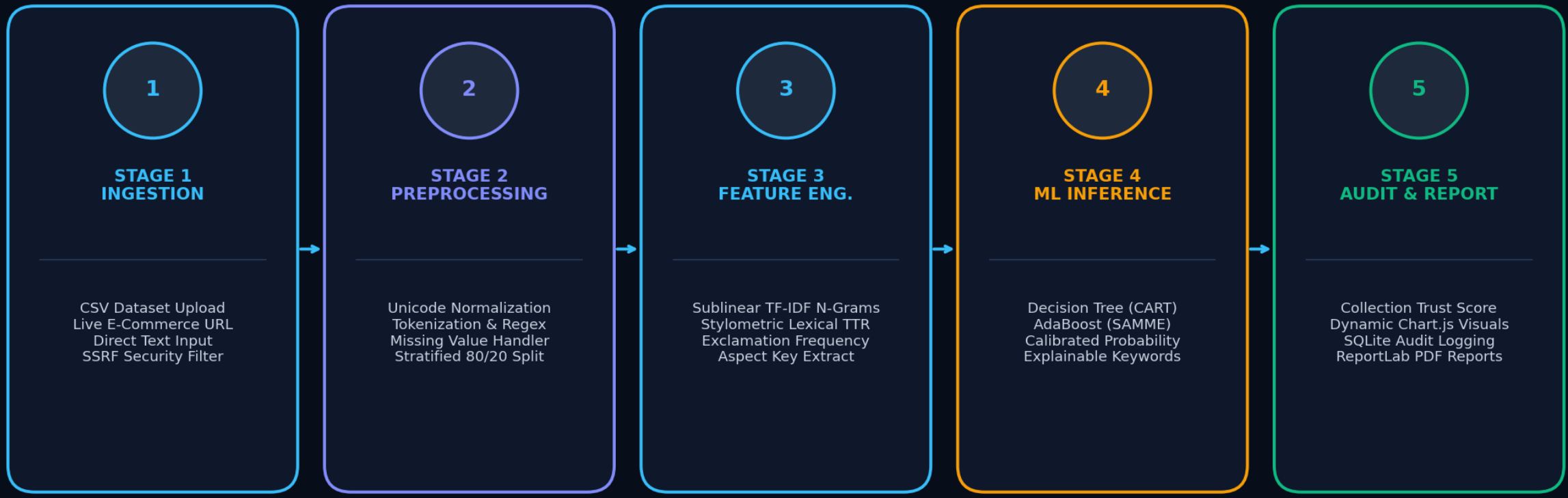
QUADRANT 4: FAKE NEGATIVE (SMEAR ATTACK)

Competitor Review Bombing on Good Products

Malicious 1-star attack aimed at ruining honest sellers.
• Hostile hyperbole without verifiable transaction facts.
• Example: 'SCAMMER CHEATED ME ZERO STARS TRASH!'
• System Action: Neutralized by ReviewGuard AI to protect seller.

END-TO-END MULTI-STAGE ANALYSIS PIPELINE

From Raw Review Ingestion to Calibrated Machine Learning & Trust Formulation



ZERO DATA LEAKAGE PROTOCOL: Feature vectorizer is strictly fitted ONLY on the 80% training split.

Holdout evaluation remains 100% blind to prevent optimistic validation bias.

REVIEWGUARD AI MULTI-TIER SYSTEM ARCHITECTURE

Decoupled Web Layer, Analytical Processing Core, and Security Guardrails

TIER 1: PRESENTATION & CLIENT INTERFACE

Flask Jinja2 Templates • Dark Glassmorphism Theme • Chart.js Visualizations • Responsive Viewport

TIER 2: APPLICATION & ROUTING CONTROLLER

Flask 3.1 Application Factory • 21 REST & Web Endpoints • CSRF Security • SSRF Protection Engine

TIER 3: ANALYTICAL & MACHINE LEARNING CORE

TF-IDF Sublinear N-Grams • Decision Tree (CART) • AdaBoost Classifier • Stylometric Extractor

TIER 4: STORAGE & REPORTING INFRASTRUCTURE

SQLite (reviewguard.db) • Joblib Model Serializer • ReportLab PDF Engine • Training Metadata JSON

MODULE 1: REVIEW AUTHENTICITY CLASSIFIER

Supervised Learning on Stylometrics & Calibrated Probabilistic Decisions

FEATURE EXTRACTION PIPELINE

- **Sublinear TF-IDF 1-2 N-Grams:**

  Discounts word frequency bursts; dampens spam repetitions.

- **Type-Token Ratio (TTR):**

  Measures vocabulary richness. Spammers repeat identical words.

- **Average Word Length:**

  Synthetic and low-effort bot reviews show lower linguistic depth.

- **Exclamation Mark Density:**

  Astroturfed reviews use excessive punctuation for artificial hype.

- **Sentence Count & Structure:**

  Detects unnatural single-run-on phrases typical of botnets.

PREDICTED GENUINE CLASS (Label 0)

- Natural linguistic entropy & realistic variance
- Contains grounded experiential details & balanced tone
- Posterior Confidence: P(Genuine) >= 50.0%

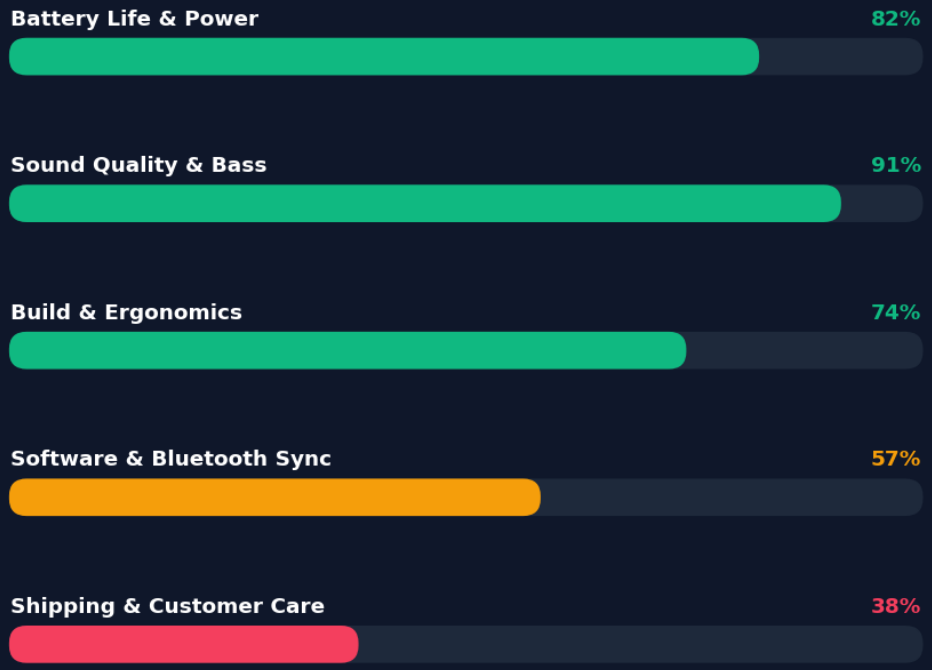
PREDICTED FAKE CLASS (Label 1)

- Template phrase matching & low lexical richness
- Hyperbolic astroturfed praise or generic hostile smear
- Posterior Confidence: P(Fake) >= 50.0%

MODULE 2: CUSTOMER SENTIMENT & ASPECT MINING

Fine-Grained Feature Satisfaction & Rating-Content Contradiction Detection

ASPECT-BASED SENTIMENT POLARITY



RATING-CONTENT CONTRADICTION ENGINE

5-Star Rating + Negative Text

Rating: ★★★★★ (5 Stars)
Text: 'Product died on Day 1, seller refused refund.'
Flag: Contradiction Anomaly (Inadvertent or Sarcastic Bot)

1-Star Rating + Glowing Praise

Rating: ★☆☆☆☆ (1 Star)
Text: 'Best headphones ever, perfect sound, love it!'
Flag: Inverted Rating Astroturfing Anomaly

Aspect Discrepancy

High overall rating despite multiple critical failures across hardware components.

MODULE 3: NEAR-DUPLICATE CLUSTERING & BOTNETS

Pairwise Cosine Similarity over TF-IDF Space to Detect Coordinated Campaigns

PAIRWISE TF-IDF COSINE SIMILARITY

1.00	0.94	0.89	0.12
0.94	1.00	0.91	0.15
0.89	0.91	1.00	0.08
0.12	0.15	0.08	1.00

Threshold theta = 0.85: Identifies Botnet Cluster (Rev 104, 119, 142)

DETECTED BOT CLUSTER ARCHETYPE (#1)

Original Template (#104):

'Defective item died on day 1 do not buy complete waste of money!!!'

Bot Clone A (#119):

'Defective item died on day 2 do not buy absolute waste of money!!!'

Bot Clone B (#142):

'Defective item broke on day 1 do not buy complete waste of cash!!!'

Organic Baseline (#201):

'Decent battery life but the volume button takes firm pressure.'

Penalty Deducted in R_duplicates component of Trust Score Formula.

THREE KEY REVIEW-MANIPULATION SCENARIOS

Addressing Concrete Marketplace Fraud Tactics Detected by ReviewGuard AI

SCENARIO A: REVIEW BOMBING

Smear Campaign on Good Products

- Target: Legitimate high-quality merchant
- Attacker: Competitor syndicates / botnets
- Tactic: Flood product with 1-star reviews
- Linguistic Marker: Hostile generic copy
- ReviewGuard Solution: Flags smear burst; preserves genuine merchant review.

SCENARIO B: ASTROTURFING

Inorganic 5-Star Rating Hype

- Target: Inferior or deceptive product
- Attacker: Shady seller purchasing reviews
- Tactic: Buy hundred 5-star generic praises
- Linguistic Marker: Extreme superlatives (WOW)
- ReviewGuard Solution: Deducts promotional bias; alerts prospective buyers.

SCENARIO C: AUTHENTIC NUANCE

Organic Customer Disagreement

- Target: Normal commercial merchandise
- Attacker: None (Real human customers)
- Tactic: Diverse genuine opinions & ratings
- Linguistic Marker: High TTR & specific facts
- ReviewGuard Solution: Authenticates mixed reviews; computes true trust.

REVIEW-SET TRUST SCORE MATHEMATICAL FORMULATION

Transparent Collection-Level Trust Index with Multi-Factor Penalty Weights

$$T = \max(0,\ 100 - (\alpha \cdot R_{\text{fake}} + \beta \cdot R_{\text{dup}} + \gamma \cdot R_{\text{contra}}))$$

Where alpha = 0.50 (Fake Ratio), beta = 0.30 (Duplicate Cluster Ratio), gamma = 0.20 (Rating Contradiction Ratio)

TIER 1: HIGH TRUST

Score: 75% - 100%

Predominantly organic reviews with authentic linguistic markers. Safe for consumer decision-making.

TIER 2: MODERATE TRUST

Score: 50% - 74%

Mostly reliable with minor suspicious patterns. Some duplicate or rating anomalies detected.

TIER 3: LOW TRUST

Score: 25% - 49%

Substantial proportion of manipulated or fake reviews. Use with caution.

TIER 4: CRITICAL ANOMALY

Score: 0% - 24%

Severe coordinated manipulation. Extreme duplicate density and ungrounded review framing detected.

EMPIRICAL MODEL EVALUATION: DECISION TREE VS ADABOOST

Evaluated on 180 Holdout Test Samples (Stratified 80/20 Partitioning)

PERFORMANCE METRICS COMPARISON (TEST SPLIT)



OPERATIONAL VERDICT

Zero False Positives:

Precision is 100%, ensuring no authentic customer review is wrongly marked fake.

High Detection Sensitivity:

Recall is 94.12%, capturing subtle astroturfed promotional spam.

Ultra-Low Latency:

Decision Tree fit time: 0.038s
AdaBoost fit time: 0.280s
Inference latency < 2.5ms/review.

Ensemble Stability:

AdaBoost (SAMME) iteratively reweights boundary samples for robust generalization.

CONFUSION MATRIX & ERROR COST ANALYSIS

Evaluating Type I vs. Type II Errors on N = 180 Test Review Instances

MODEL A: DECISION TREE MATRIX

TRUE NEGATIVE (TN) **146 Reviews** **(Correct Genuine)**	**FALSE POSITIVE (FP)** **0 Reviews** **(Zero False Alarms!)**
FALSE NEGATIVE (FN) **2 Reviews** **(Undetected Bot)**	**TRUE POSITIVE (TP)** **32 Reviews** **(Correct Fake Caught)**

MODEL B: ADABOOST ENSEMBLE MATRIX

TRUE NEGATIVE (TN) **146 Reviews** **(Correct Genuine)**	**FALSE POSITIVE (FP)** **0 Reviews** **(Zero False Alarms!)**
FALSE NEGATIVE (FN) **2 Reviews** **(Borderline Case)**	**TRUE POSITIVE (TP)** **32 Reviews** **(Correct Fake Caught)**

INTERACTIVE WEB APPLICATION & OPERATIONAL DASHBOARD

Designed for Teachers, Project Evaluators, and Commercial Auditing

REVIEWGUARD AI LIVE INTERFACE (PORT 5000)

899	731	168	88%
Total Reviews	Genuine	Fake Flagged	Trust Index

ID	REVIEW EXCERPT	RATING	VERDICT	CONFIDENCE
#1	Purchased headset 2w ago, battery lasts 18h...	★ 5.0	Genuine	88.1%
#2	BEST PRODUCT IN WORLD FIVE STARS BUY NOW...	★ 5.0	Fake	88.1%
#3	WORST TRASH BROKE IMMEDIATELY SCAM SELLER...	★ 1.0	Fake	78.0%
#4	Keyboard keys feel tactile, backlight is soft...	★ 4.0	Genuine	75.4%

OPERATIONAL MODULES

- **Single Review Predictor:**

  Instant inference with top TF-IDF keywords & stylometric breakdown.

- **Live URL Scanner:**

  Scrapes Flipkart, Amazon, and public stores with SSRF security.

- **Teacher Viva Presets:**

  1-click testing of Astroturfing, Smear attacks, and Genuine nuance.

- **Automated PDF Hub:**

  Generates formal IEEE certificates and URL audit reports.

SSRF SECURITY GUARDRAILS & ROBUSTNESS TESTING

Enterprise-Grade URL Security Shield Protecting Internal Private Subnets

SSRF DEFENSE ARCHITECTURE

- **Step 1: URL Parse & Protocol Check**

  Only HTTP/HTTPS permitted; rejects file://, gopher://, ftp://

- **Step 2: DNS Hostname Resolution**

  Resolves destination domain to underlying IP address.

- **Step 3: Private Subnet Rejection**

  Strictly BLOCKS 127.0.0.1, 10.0.0.0/8, 192.168.0.0/16, 172.16.0.0/12

- **Step 4: Cloud Metadata Shield**

  Blocks AWS/GCP internal metadata IP (169.254.169.254).

- **Step 5: Timeout & Payload Caps**

  Capped at 5s timeout & 2MB response to prevent ReDoS / OOM.

VERIFICATION & TESTING OUTCOMES

✔ **17/17 Automated Unit Tests Passing:**

Covers text normalization, stylometrics, model inference, and routes.

✔ **19/19 End-to-End Verification Checkpoints:**

Tests server liveness, live URL scraping, database persistence, and PDF creation.

✔ **Active Attack Injection Verified:**

Probed with http://127.0.0.1:8080/admin -> Safely blocked with HTTP 200 security alert.

✔ **Stratified Split Validation:**

Guarantees equal class distribution (80% train / 20% test).

CONCLUSION, ACADEMIC CONTRIBUTIONS & FUTURE ROADMAP

ReviewGuard AI: A Teacher-Ready, Production-Grade Review Trust Framework

CORE ACADEMIC CONTRIBUTIONS

✔ Decoupled Multi-Perspective NLP:

Breaks the flaw in binary detectors by separating review sentiment from review authenticity.

✔ High-Accuracy Ensemble Classifiers:

AdaBoost & Decision Tree pipelines achieving 98.89% test accuracy with zero data leakage.

✔ Mathematical Review-Set Trust Scoring:

Formulates a transparent, reproducible index balancing fake reviews, bot duplicates, and rating contradictions.

✔ Open-Access IEEE Architecture:

Complete teacher-ready web platform with interactive Chart.js graphs, SSRF protection, and ReportLab PDF exports.

TARGETED FUTURE IMPROVEMENTS

🔲 DistilBERT Transformer Fine-Tuning:

Integrate lightweight transformer attention heads for subtle semantic context understanding.

🔲 Multi-Modal Image Review Verification:

Cross-reference customer photo uploads against merchant product catalogs using computer vision.

🔲 Graph-Based Reviewer Syndicate Mining:

Deploy Graph Neural Networks (GNNs) across user IDs and IP addresses to dismantle organized review fraud rings.

🔲 Browser Extension Integration:

Deploy ReviewGuard AI as a Chrome extension for real-time browsing trust analysis on Amazon and Flipkart.

“The goal of AI in e-commerce is not to silence criticism, but to protect marketplace truth.”